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Incident Timeline and Evidence Collection for Small Teams

Team TriCipher

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Problem Description

During a cybersecurity attack, small security teams often struggle to understand what happened because logs are scattered across different systems and use different timestamp formats and time zones.

- Manually collecting and analyzing logs is time-consuming.
- Different timestamps make it difficult to arrange events in the correct order.
- Original evidence may be accidentally modified or lost.
- Missing or incorrectly formatted log entries can hide important information.
- Creating a clear incident timeline and report requires significant manual effort.

Background

Cybersecurity incident investigation involves analyzing system and application logs to understand how an attack occurred, identify suspicious activities and reconstruct the sequence of events.

However, logs collected from different sources often have different formats, time zones and clock settings, making it difficult to establish an accurate incident timeline.

Security Operations Centers (SOCs) use log monitoring, event correlation and incident triage to identify threats and support timely incident investigation .

Literature & RFC Review

- ED-ASOC [1]: Event-driven ingestion, RAG-based investigation and human-supervised response. Limitation: Scalability and adversarial robustness.
- AgentSOC [2]: Multi-layer alert analysis, validation and risk-aware response. Limitation: Limited-scale evaluation.
- ASOC-CA [3]: Automates alert classification, MITRE ATT&CK mapping and incident reporting. Limitation: Limited synthetic attack scenarios.
- Evolution of Agentic AI [4]: Reviews the progression from single-LLM systems to multi-agent cybersecurity pipelines.

Research Gap: Existing approaches address individual SOC tasks, highlighting the need for an integrated multi-agent workflow combining alert triage, contextual investigation, response orchestration and evaluation, alongside evidence preservation and incident timeline reconstruction.

Proposed Solution

- Log Ingestion: Collect logs and alerts from Zeek and Elastic SIEM.
- Evidence Preservation: Retain original logs and verify integrity using SHA-256 hashes.
- Timestamp Normalization: Convert timestamps to UTC and account for known clock skew.
- Incident Timeline: Create a searchable, chronological view of security events.
- Multi-Agent Investigation: Use LangGraph agents for triage, investigation and response recommendations.
- Evaluation & Reporting: Identify parsing gaps, evaluate results and generate incident reports.
- Visualization: Integrate SQLite, Grafana and the SOC Copilot for monitoring and investigation.

Optimization Proposed

Before (Existing SOC)

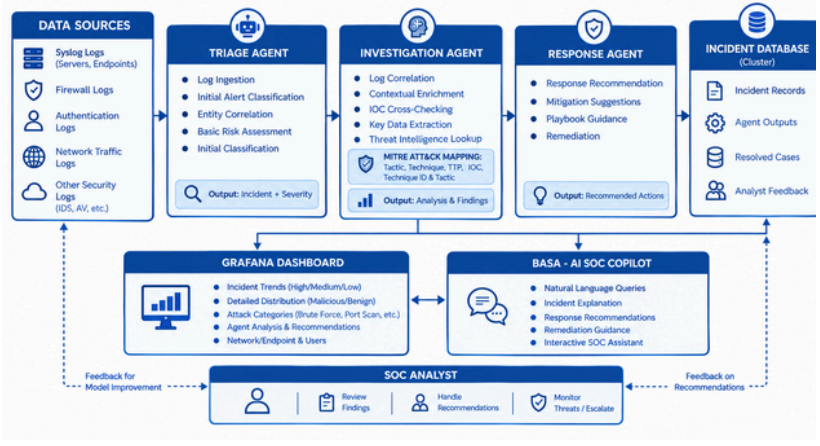
- Logs and alerts analyzed through the SOC pipeline
- No dedicated evidence integrityverification
- Timestamps may use different time zones
- Incident investigation through multi-agent workflow
- Limited visibility into parsing failures
- Incident information stored and visualized
- Investigation efficiency not formally benchmarked

After (Proposed Enhancement)

- Logs collected from multiple sources and formats
- Original logs preserved and SHA-256 hashes generated
- Timestamps normalized to UTC with known clock skew.
- Searchable chronological incident timeline
- Parsing gaps and malformed records documented
- Structured incident reports with timeline and evidence details
- Triage time compared against a manual baseline

Solution Architecture & Design

Visual: Flowchart or diagram showing the optimization process.



Data Flow Diagram

Overview of the architecture. Include the flow chart diagram created earlier. Visual: Diagrams or architecture visuals.



Deployment Timeline

Phase 1: Oct 15–20 — Core Feature Development

- Enhance the existing SOC pipeline.
- Develop evidence preservation and timestamp handling.

Phase 2: Oct 21–26 — System Integration

- Integrate incident timeline reconstruction.
- Extend investigation and reporting capabilities.

Phase 3: Oct 27–31 — Testing & Finalization

- Validate system functionality and reliability.
- Evaluate performance and prepare the final demo.

Bibliography I

[1] "Event-Driven Agentic SOC (ED-ASOC): Supervisor-Based LLM Framework for Dynamic Incident Response and SOAR Orchestration." https://ieeexplore.ieee.org/abstract/document/11458922/

[2] "AgentSOC: A Multi-Layer Agentic AI Framework for Security Operations Automation." https://ieeexplore.ieee.org/abstract/document/11395783/

[3] "AutoSOC Cyber Analyst (ASOC-CA): Using AI to Automate SOC Tier 1 & 2 Activities." https://ieeexplore.ieee.org/abstract/document/11393893/

[4] "The Evolution of Agentic AI in Cybersecurity: From Single LLM Reasoners to Multi-Agent Systems and Autonomous Pipelines." https://ieeexplore.ieee.org/abstract/document/11395809/

Thank You for Your Attention!